

EcoLens

A Chrome Extension That Makes the Carbon Cost of Everyday Browsing Visible in Real Time

Parul University Environment Hackathon 2026 | Theme: Designing Sustainable Futures in a Digital Age

1. Project Overview and Introduction

Project Title	EcoLens — Real-Time Digital Carbon Footprint Tracker (Chrome Extension)
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University	Parul University
SDG Alignment	SDG 12 (Responsible Consumption & Production), SDG 13 (Climate Action), SDG 9 (Industry, Innovation & Infrastructure)

EcoLens is a Manifest V3 Chrome extension that makes the carbon cost of everyday browsing visible in real time. It tracks user activity across nine supported platforms — Google Search, ChatGPT, Claude, Gemini, Perplexity, Netflix, YouTube, and Spotify — and converts that activity into estimated CO2 emissions using a four-factor model: network data transfer, device energy consumption, live grid carbon intensity, and site-specific baselines. Results are surfaced through an in-page badge injected onto each tracked site and a full popup dashboard, ensuring the environmental cost of each digital action is always within the user's view.

This project addresses one of the least-discussed sustainability frontiers: the hidden carbon footprint of routine digital behaviour. The digital sector contributes approximately 3–4% of global greenhouse gas emissions — a share growing faster than aviation — yet individual users receive no environmental feedback at the point of consumption. EcoLens closes this gap by embedding sustainability awareness directly into the browser. It aligns with SDG 12 by making consumption costs transparent, SDG 13 by enabling climate-informed choices in real time, and SDG 9 by demonstrating that responsible innovation can be privacy-preserving and zero-infrastructure by design.

2. Problem Statement and Objectives

A single Google search generates roughly 0.3g CO2; one ChatGPT prompt produces approximately 3g — ten times more. An hour of Netflix streaming consumes around 0.1 kWh. These figures are well-documented in peer-reviewed literature, yet they are entirely invisible to the person generating them. Unlike physical consumption — where energy labels, recycling infrastructure, and carbon-offset prompts are commonplace — digital consumption provides zero environmental feedback at the moment of action.

This is not a values gap. Surveys consistently show that internet users care about climate change, but have no mechanism to connect their daily browsing habits to real-world emissions. EcoLens addresses this as a precise information-delivery problem: the data exists in research literature, but has never been packaged into a consumer-facing, real-time, per-action format available at the exact moment a user makes a digital choice.

Objectives

- Display a real-time CO2 estimate for each digital action at the moment it occurs, without interrupting the user's workflow.

- Model emissions accurately using a four-factor approach: network transfer, device energy, live grid carbon intensity, and site-specific baselines.
- Detect AI model variants (GPT-4o, Claude Sonnet, Gemini Pro, etc.) to provide granular, model-level emissions breakdowns.
- Store a transparent provenance record for every tracked event so users can inspect exactly how each estimate was calculated.
- Nudge users toward lower-carbon alternatives through contextual tips, daily carbon budgets, and 7-day and 30-day trend insights.

3. Proposed Solution

EcoLens injects a content script into each of its nine supported platforms. The script identifies the action type — a search query, an AI prompt submission, or active media playback — and measures network bytes transferred via Chrome's webRequest API. This is combined with device energy heuristics (derived from battery data or a user-selected device profile: phone, laptop, desktop, or TV) and live grid carbon intensity from the Electricity Maps API. When live data is unavailable, the extension falls back to regional carbon intensity averages based on detected country. The final CO2-equivalent estimate is stored per site, per day, and per AI model in chrome.storage.local — entirely on-device with no external server required for core functionality.

Two interfaces deliver results to the user:

- **In-page Badge:** injected directly onto the tracked site, showing live CO2 estimate, grid intensity and region, data transferred, detected AI model, streaming watch time, and a real-world equivalence comparison benchmarked against Netflix-per-hour.
- **Popup Dashboard:** a full summary including today's total emissions, per-platform breakdown, 7-day and 30-day trend charts, model-level AI breakdown, provenance labels (measured vs. estimated), daily budget alerts at 80% and 100%, personalised reduction tips, and a device selector.

Technology Stack

Component	Implementation
Extension Architecture	Chrome Manifest V3 with Service Worker background script; no build step required
Activity Tracking	Content scripts with MutationObserver for single-page app route changes
Network Measurement	chrome.webRequest API; falls back to site-specific baseline estimates
Grid Carbon Intensity	Electricity Maps API (live); regional average fallback by detected country
Device Energy	Battery-derived heuristics or user-selected profile (phone / laptop / desktop / TV)
AI Model Detection	Heuristic UI-label detection: GPT-3.5 to o3, Claude Haiku/Sonnet/Opus, Gemini Flash/Pro/Ultra, Perplexity Sonar/Pro/Reasoning
Data Storage	chrome.storage.local — 180-day daily totals, 90-day activity events; fully on-device
Optional Cloud Sync	Supabase backend for sign-in, 30-day sync, social challenges, and leaderboards

A distinguishing innovation is EcoLens's trust and provenance layer. Every tracked event stores its measurementMode (measured or estimated), gridSource (live, fallback, or default), modelConfidence (detected or defaulted), and a full energy breakdown across network, baseline, and device components. The

popup uses this data to explain precisely where each estimate came from — making EcoLens uniquely auditable and building genuine user trust rather than presenting opaque carbon numbers.

4. Results and Feasibility

EcoLens operates at the behavioural layer of sustainability — the layer where individual awareness translates into collective action. Research in behavioral economics consistently shows that real-time, contextual feedback is among the most effective mechanisms for shifting habitual behaviour: smart electricity meters reduce household energy consumption by 5–15% through awareness alone, without requiring any change in lifestyle or infrastructure. EcoLens applies this proven principle to digital consumption across nine of the world's most-visited platforms, targeting habits that recur dozens of times every day.

The core extension requires no backend infrastructure, eliminating hosting costs and data-privacy concerns entirely. All tracking, estimation, and storage happens on-device. The working prototype communicates its value in under three seconds — open a browser, perform a Google Search, and the CO2 badge appears immediately — an unusual advantage in a competition setting where most projects require lengthy explanation before a judge can evaluate them. The codebase can be published to Chrome Web Store, Firefox Add-ons, and Microsoft Edge Add-ons with minimal adaptation, meaning a single team's work can reach billions of users at near-zero marginal cost.

Feasibility Assessment

Dimension	Assessment
Technical	Manifest V3 is stable and well-documented. Full feature set is achievable by a two-person team with no build step — the extension runs directly from source files.
Data Reliability	Estimates combine peer-reviewed baselines (Aslan et al., IEA, Shift Project) with live grid data from Electricity Maps. Provenance labels distinguish measured from estimated values for every event.
Privacy & Ethics	Fully local-first. No browsing data, search queries, or identifiers are transmitted without explicit user opt-in to the optional cloud sync feature.
Scalability	Publishable to Chrome, Firefox, and Edge stores. Optional Supabase backend enables social challenges, team leaderboards, and institutional dashboards as usage grows.

Future Developments

- Live model-energy API integration to replace heuristic AI estimates with provider-reported usage data as major AI companies begin publishing per-query energy disclosures.
- Institutional dashboard allowing universities and organisations to aggregate anonymised, opt-in data and set collective digital carbon reduction targets.
- Firefox and Safari ports to extend coverage to all major browsers and reach users beyond the Chromium ecosystem.

References

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